

CAD in Robotics

16328 Trial N' Error

Introduction

What is CAD used for?

- helps users create designs in either 2D or 3D models to visualize construction, and enables the development, modification, and optimization of the design process.
- In Robotics we can use CAD to speed up/aid the modification or creation of the different systems on our robot.

What CAD software should I use?

- The CAD software you use is an important decision that can dictate many factors from your workflow as a team or the complexity of the parts you create.
- You should choose the CAD software that best matches your current ability, or what you are willing to put effort into learning.
- Solidworks
 - Pros:
 - Large software library including manufacturing tools, motion study, and online/desktop applications
 - Capable software with tools to aid in any kind of design.
 - Large quantity of tutorials and guides
 - Cons:
 - Price (can be acquired through [sponsorship](#))
 - Steeper learning curve compared to other software

- OnShape
 - Pros:
 - Free to use
 - Online application with no desktop application needed
 - Active community
 - Cons:
 - Steeper learning curve compared to other software
 - Limited capabilities for simulations and motion study
- Fusion360
 - Pros:
 - Free to use
 - Full-featured desktop CAD software with motion studies and simulation tools
 - Active community
 - Cons:
 - Steeper learning curve compared to other software
 - Missing some advanced tools in the free version
- Tinkercad
 - Pros:
 - Free to use
 - Easy to learn and use
 - Cons:
 - Limited capabilities in all aspects of design
 - Can be slower than using more advanced CAD software

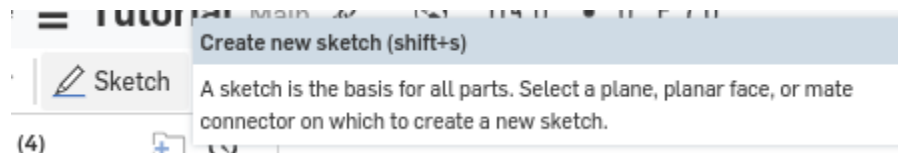
Sketches

Once you have chosen your CAD software, it is time to start designing. Lucky for you, most CAD software (assuming it's not Tinkercad) follows the same simple rules and has similar tools to create designs, making it easier to switch to more advanced software later on. For this tutorial, I used OnShape.

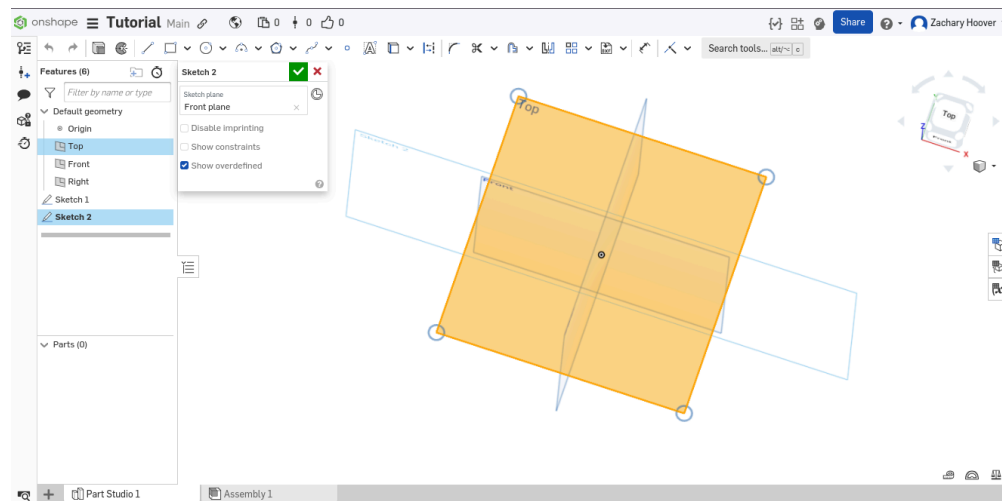
Creating Sketches

Sketches are the basis of modern CAD design, they can be edited, extruded, and can be put on any surface in your model.

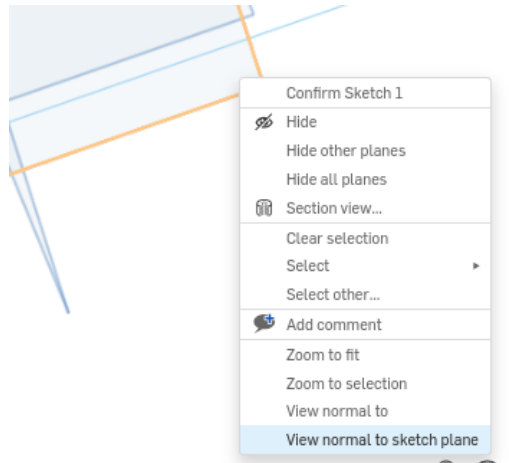
1. Click the sketch icon



2. Select the plane you want to sketch on



3. Normal to the plane by right-clicking on it and selecting the correct 'normal to sketch plane' option (some software might have an icon for this)



Drawing a Sketch

Once you create a sketch there are a variety of tools you can use to define the lines that will make up the sketch.

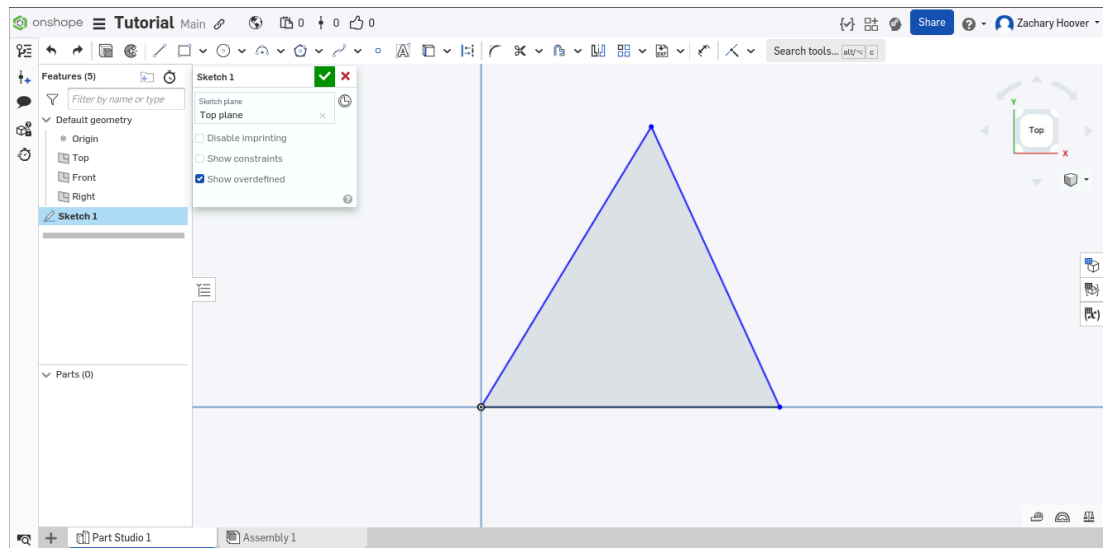
Tip 1: You can click on the corners or sides of other lines when drawing your sketch to constrain the sketch.

Tip 2: If your software supports it, you can use construction lines (sometimes called center lines) to create lines that will not be part of the final sketch, but help you more accurately place lines.

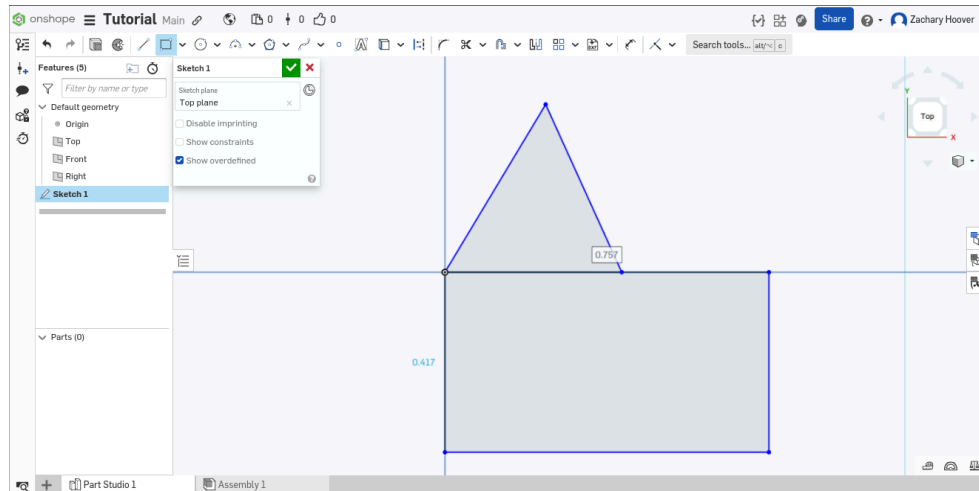
Tip 3: Make sure all areas you plan to extrude are completely closed in (usually shown by being a darker region).

Tip 4: Don't worry too much about the accuracy of your sketch just yet, you will go back later and add dimensions to constrain your sketch

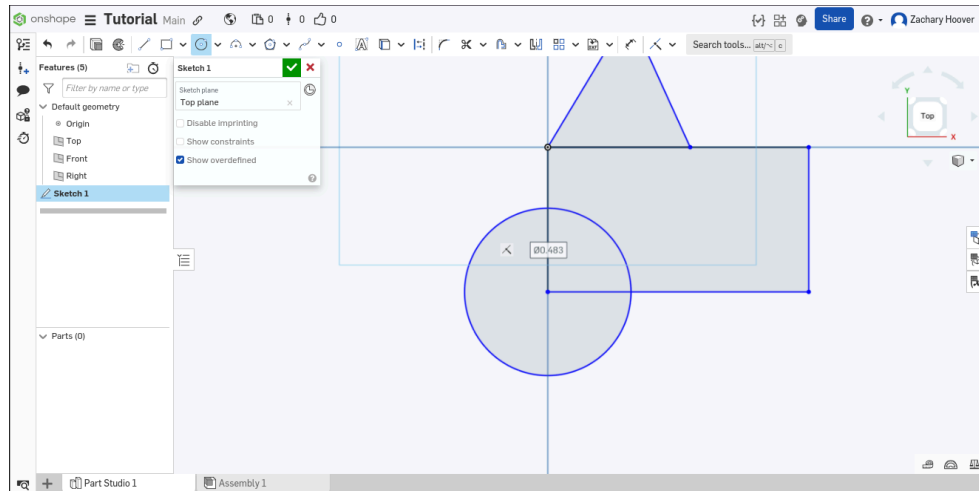
- Lines can be used to make almost any shape with segments.



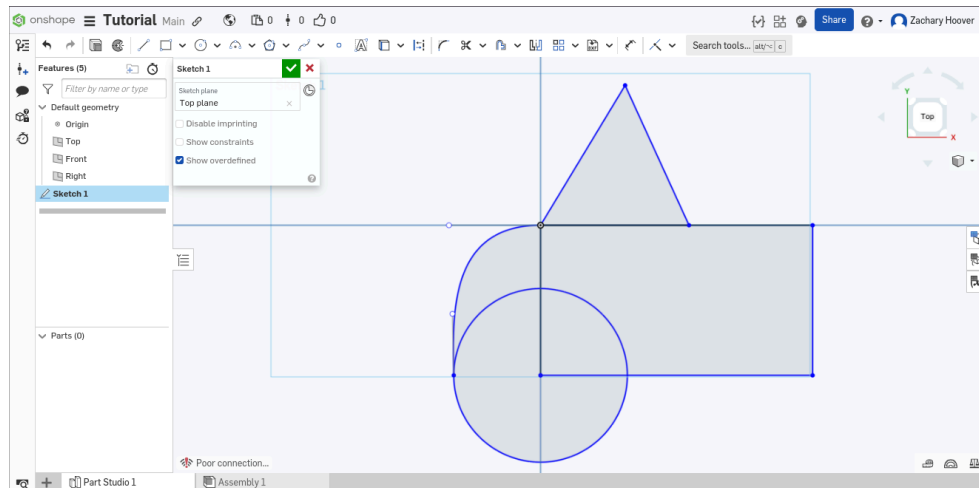
- Rectangles are exactly what they say they are, boxes with four sides and four 90-degree angles.



- Circles are exactly what they say they are, they are usually sketched from their center point.



- Splines are curved lines. The intensity of this curve is usually adjustable for applications such as ramps or edges.



Constraining a Sketch

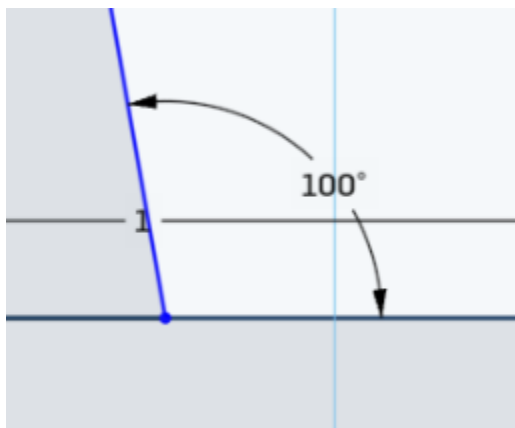
Once you have the general layout of your sketch completed, you have to constrain your sketch by adding dimensions before you can create features with it.

Tip 1: Don't move on from a sketch until it has been entirely constrained, this could lead to issues later in your design

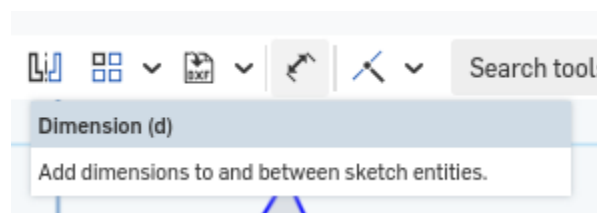
Tip 2: Most CAD software has a key bind for their constraining tool, this key bind is usually 'd'

Tip 3: A common mistake many people make is not checking the units their document is set to before creating constraints, this can end up having a huge impact on the scale of your model.

Tip 4: Some CAD software allows for the creation of angles at an intersection of two lines by selecting those lines and entering the angle in degrees



1. Select your dimension tool (or use the key-bind)

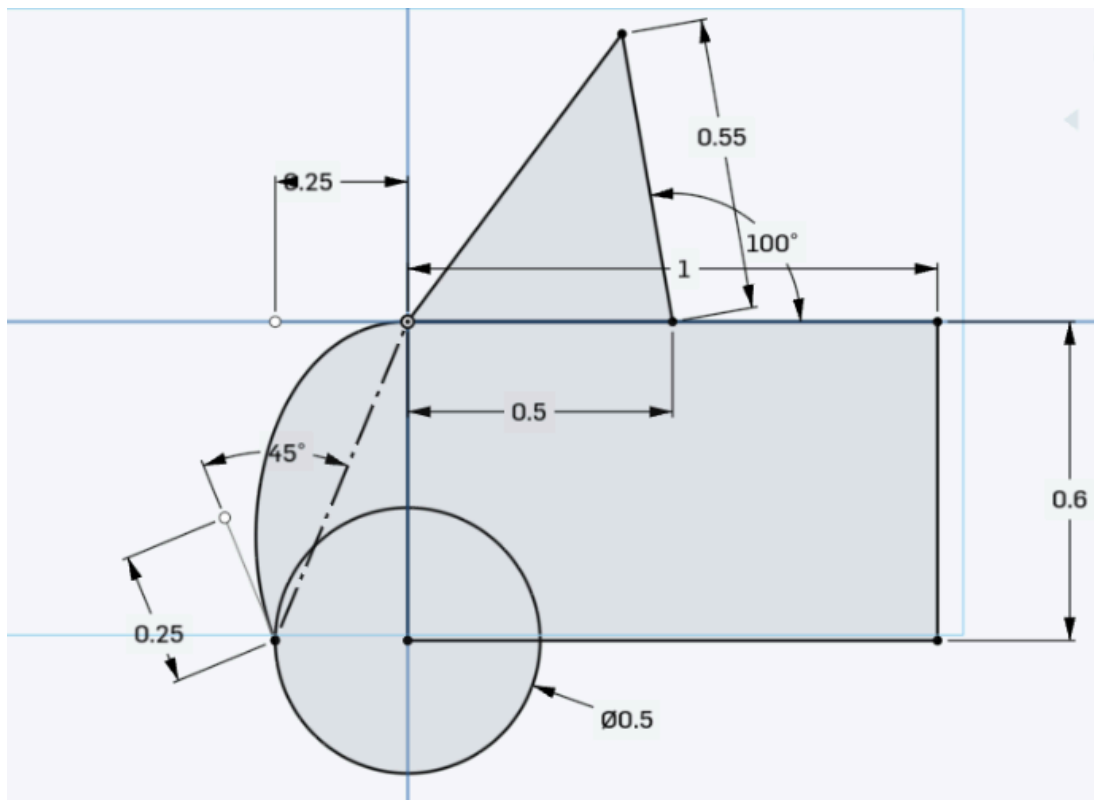


2. Click the line or sketch object or points you would like to constrain and click again to place the dimension object in a location where it's easily identifiable (some CAD software may do this automatically)



3. Enter the constraint value and press enter (can change depending on the software)

Fully Constrained example:



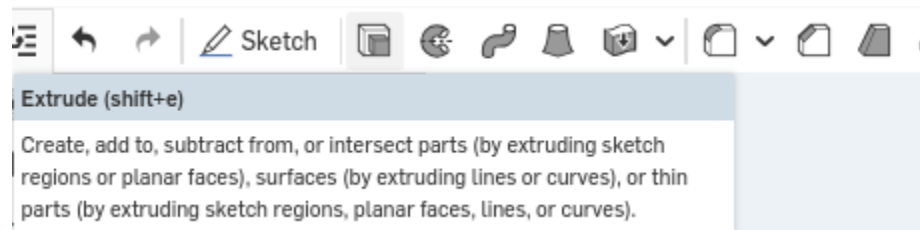
Features

Once you have your first sketch, you can create features using the different parts of the sketch. Features can vary widely from each other in what they require as parameters, and how they can change your document, so some of this information may not carry over to other CAD software

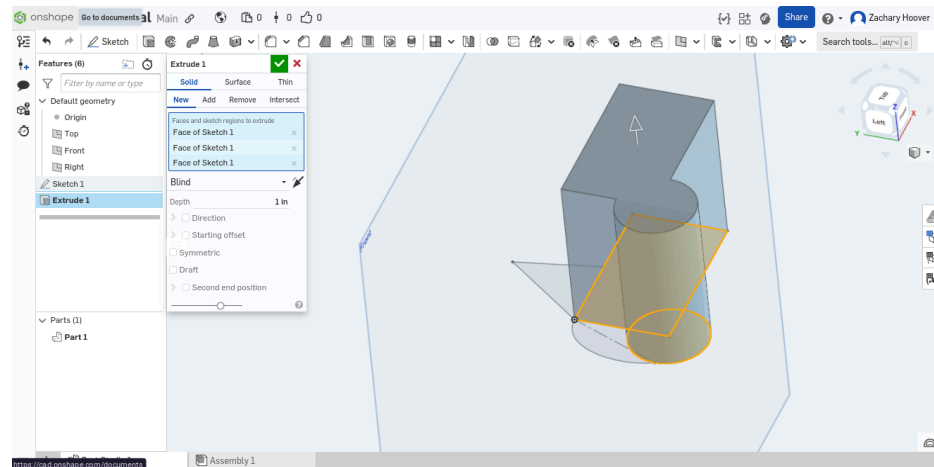
Creating Features

Tip 1: Don't worry about some of the parameters you see when creating a feature, you will learn them over time when you need them.

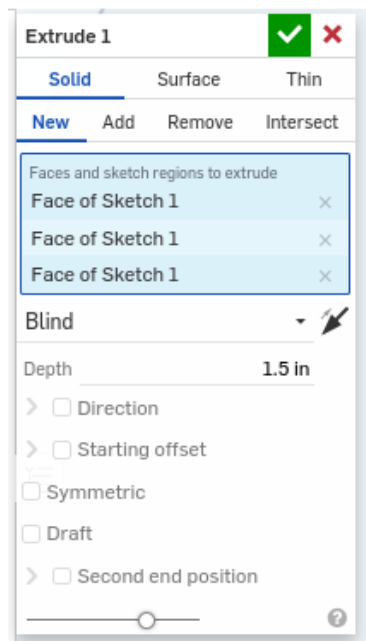
1. Select the feature (in this case, extrude) you would like to add



2. Select the parts of your sketch you would like the feature to apply to (you can have multiple features on a single sketch, but some CAD software will hide a sketch in the object tree once it's been used for a feature. When this happens, you can right-click on the object and unhide it)

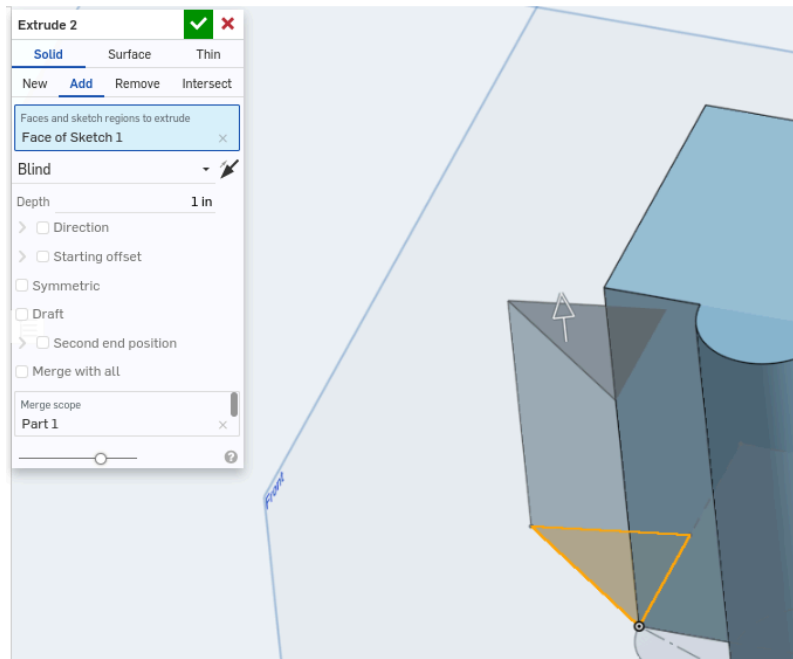


3. Fill out all required parameters for the feature (extrude amount, add/remove, pattern amount, point of rotation, etc.). Press the check when done.

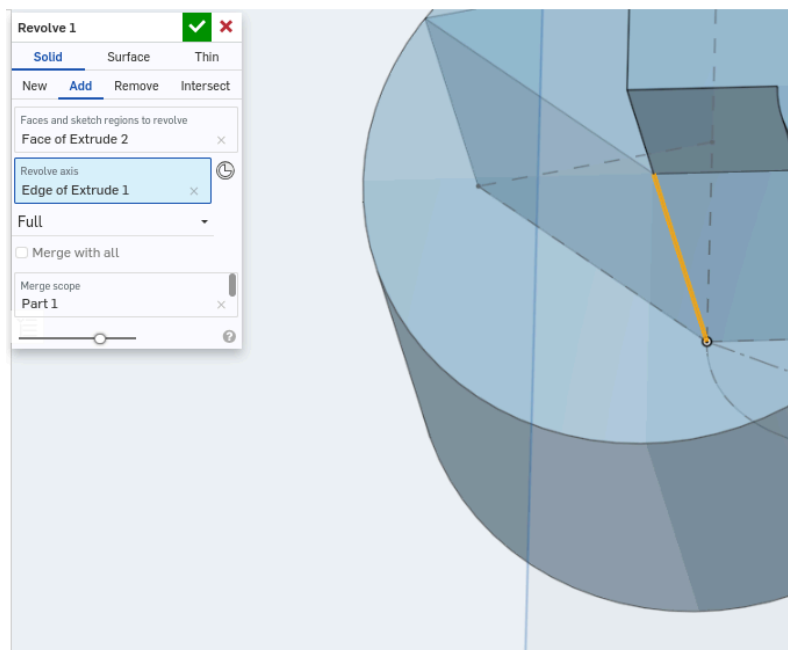


Parameter Types

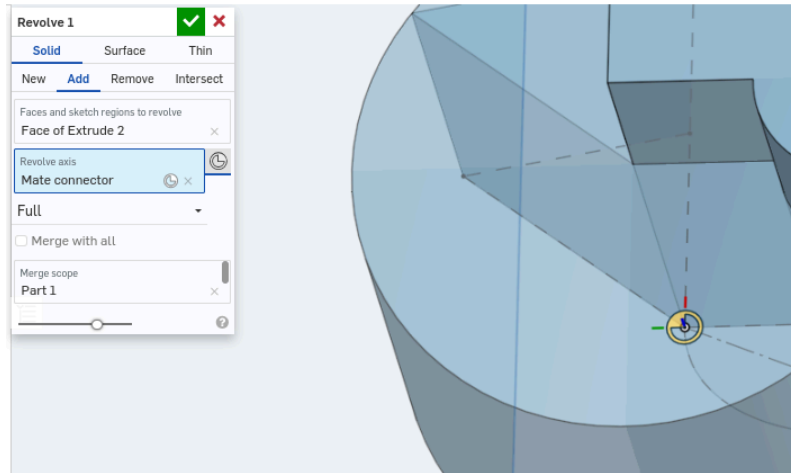
- **Surfaces, planes, or faces** are flat parts of an object or sketch. The surfaces can sometimes be used for directions, or the basis of a feature.



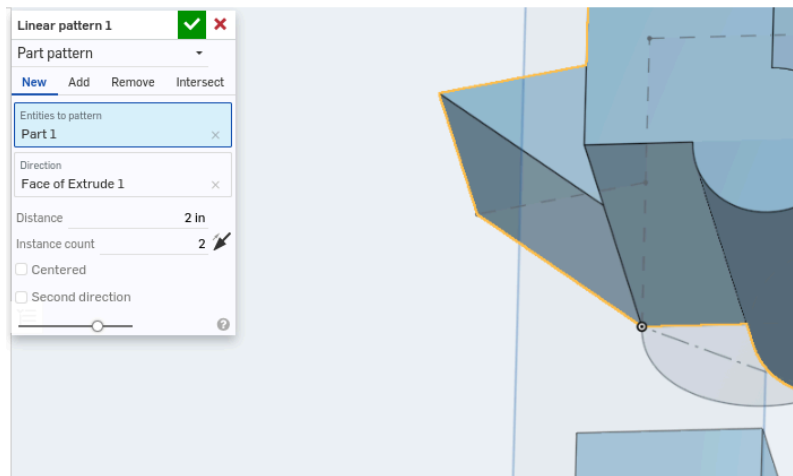
- **Edges** are at the corners of objects, and can also be lines from a sketch



- **Points or vertices** are where lines intersect, or are the center of a circle, or are any points that are added in a sketch.



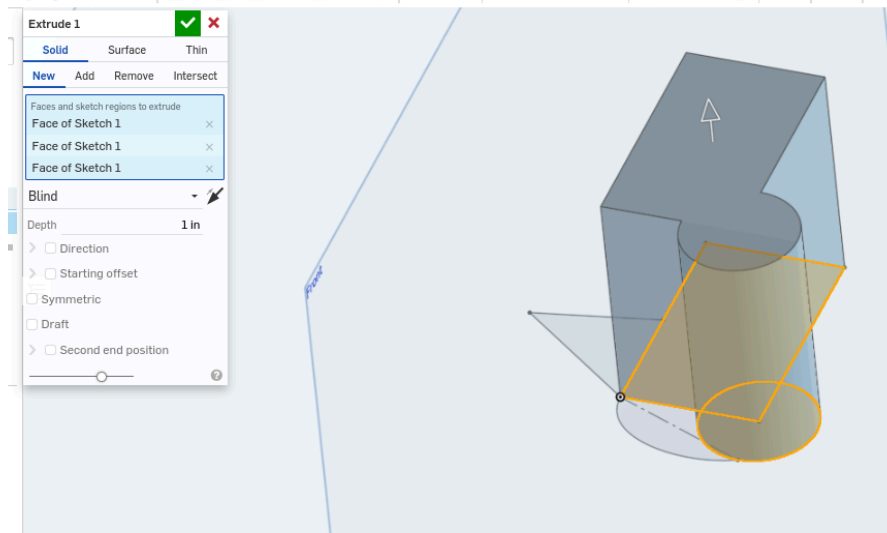
- **Parts or features** in some cases also parameters for another feature, such as when creating patterns.



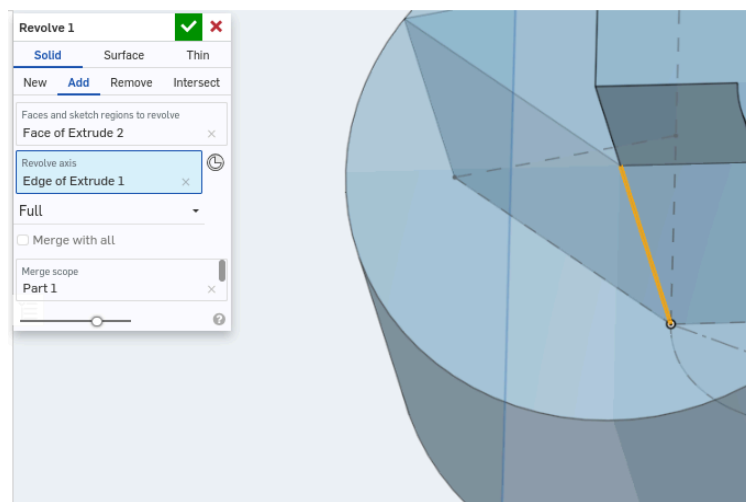
Basic Features

Tip 1: When creating features, it's best to experiment with them and learn them over time, as such only the main features will be covered in this tutorial.

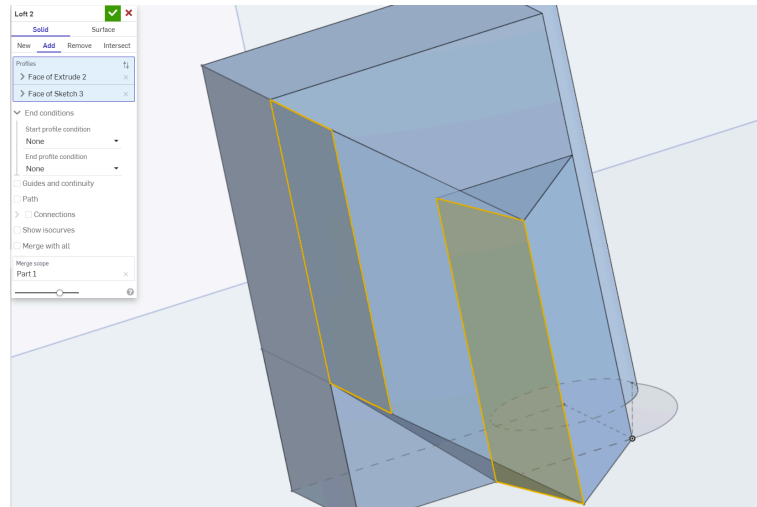
- **Extrudes** are the most common features you will create, these allow you to add or remove material linearly using sketches or surfaces.



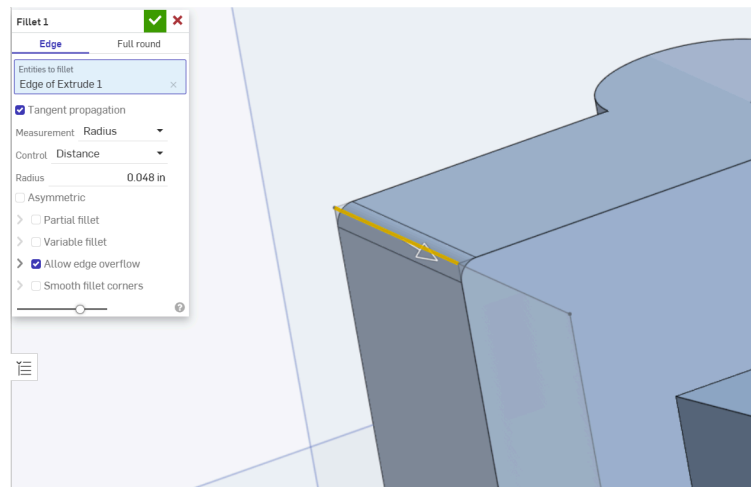
- **Revolves** allow you to extrude a surface around a point of rotation or edge



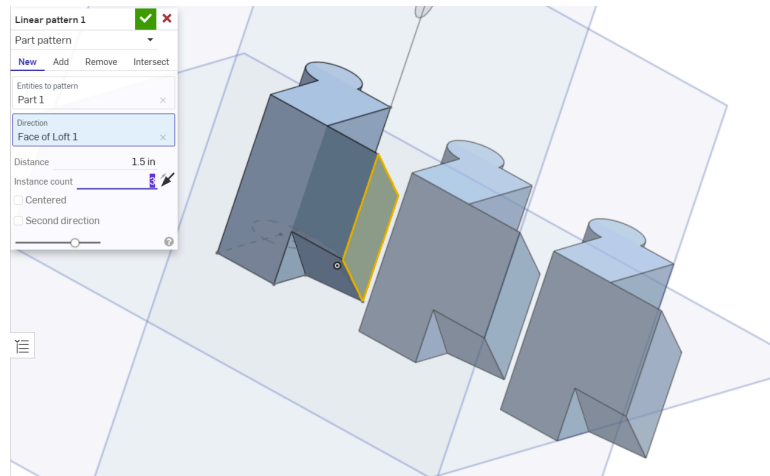
- **Lofts** allow you to extrude a surface up to another, with the ability to extrude to surfaces that are different sizes and shapes.



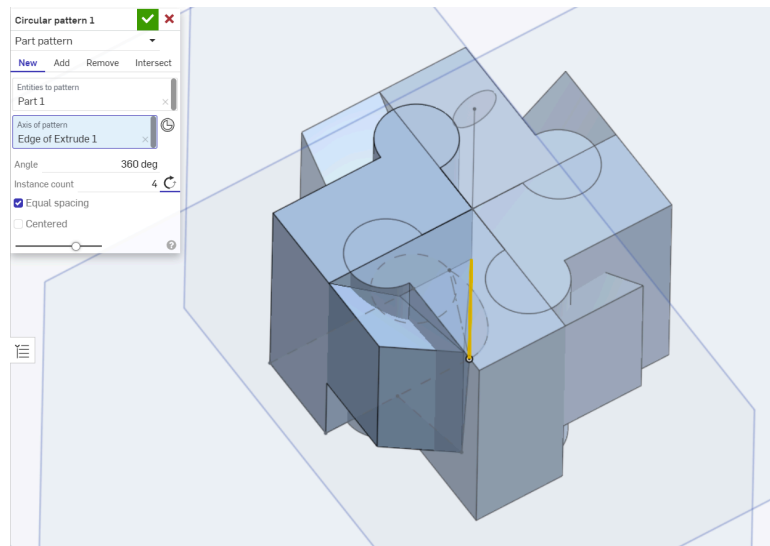
- **Filets** are used to round edges, these can be to remove sharp edges on designs, or strengthen regions with sharp edges.



- **Linear Patterns** are used to repeat features or parts in a straight line.

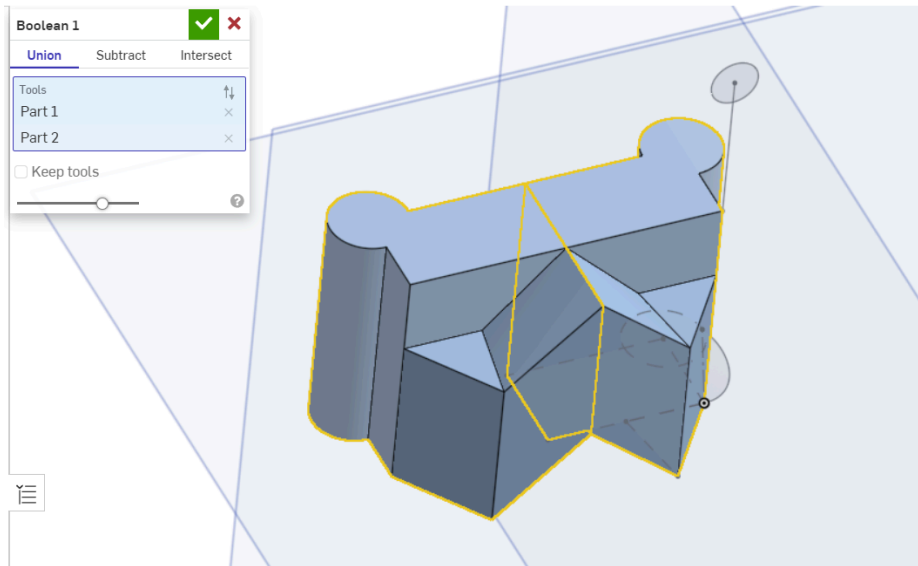


- **Circular Patterns** are used to repeat features or parts around a center point.

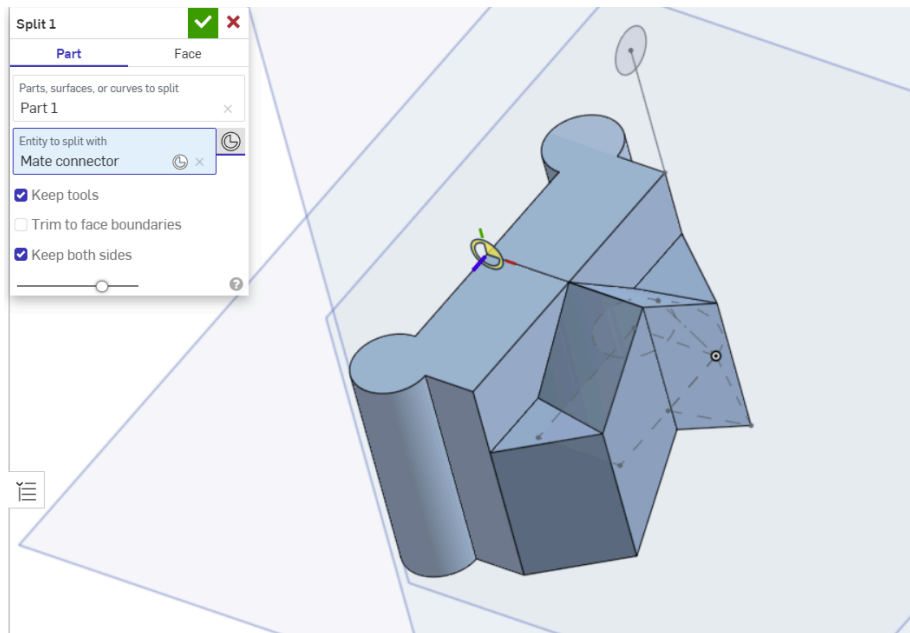


Tools

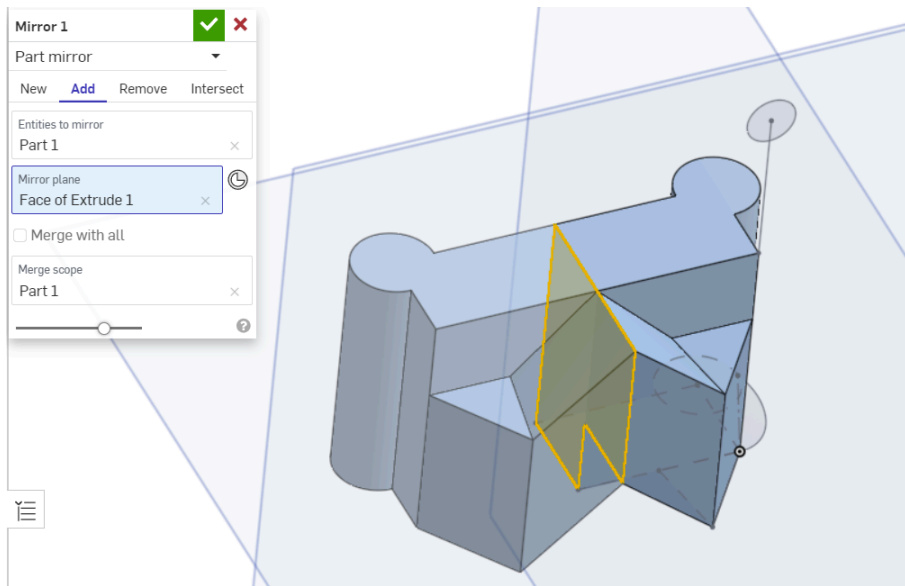
- **Booleans** are used to merge two parts into one.



- **Split** is used to split a part into two separate parts.



- **Mirror** is used to create a mirrored copy of features and parts.



Glossary

This section is for words/phrases that may be confusing.

CAD - Computer Aided Design.

Snap Point - Intersection of two lines, center of a circle, points that the mouse snaps to in CAD software. Usually used when defining sketches or defining parameters for features.

Feature - Extrusion, Fillet, Loft, Pattern, all the parts that make up a design in CAD.

Plane - A default surface you can create sketches on.